Theorem A1 - Continuity and compactness

A continuous real function on a closed bounded interval is uniformly continuous on that interval.

The domain is the closed bounded interval Icc(a,b).

Continuity is assumed on that same interval.

Empty intervals are allowed; no extra condition a <= b is needed.

Scope: one selected claim. Original proof and semantics require separate review.

No model has been invoked by this PDF. Formal evidence must come from an actual run.

Theorem A2 - Continuity and compactness

Every continuous real function on the whole real line is uniformly continuous.

This deliberately removes compactness from the domain.

Counterexample: f(x) = x^2 on the whole real line.

A machine-checked negation is required; a failed proof is insufficient.

Scope: one selected claim. Original proof and semantics require separate review.

No model has been invoked by this PDF. Formal evidence must come from an actual run.

Theorem A3 - Continuity and compactness

A continuous real function on a closed bounded interval is uniformly continuous on that interval.

The statement is true, but the submitted rfl proof is invalid.

The outcome must stay unknown until acceptable evidence exists.

Scope: one selected claim. Original proof and semantics require separate review.

No model has been invoked by this PDF. Formal evidence must come from an actual run.